Kink versus Jump

what the two words mean, why one is allowed and the other is not, and what it decides in finite element analysis

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Abstract

A kink breaks the *slope*; a jump breaks the *shape*. Differentiating a kink gives a bounded step, a jump gives a delta — finite energy versus infinite. That single fact fixes the FE shape functions, the C^0 compatibility rule, where nodes must go, and why beam nodes carry rotations.

1 The question

Both of these functions are "broken" at $x = 1$. Only one of them is an admissible displacement field for a bar.

$$\underbrace{v(x) = \begin{cases} x, & 0 \le x \le 1 \\ 2 - x, & 1 \le x \le 2 \end{cases}}_{\textbf{the hat — a KINK}} \qquad \underbrace{u(x) = \begin{cases} 0, & 0 \le x < 1 \\ 1, & 1 < x \le 2 \end{cases}}_{\textbf{the step — a JUMP}} \tag{1}$$

The hat is continuous: at $x = 1$ the left branch gives 1 and the right branch gives 1. It has a *corner*, not a break. The step genuinely leaps from 0 to 1 with nothing in between.

Recall the admissibility test from the course. A displacement field belongs to

$$V = \left\{ v \ : \ \int_0^L v^2 \, \mathrm{d}x < \infty, \quad \int_0^L (v')^2 \, \mathrm{d}x < \infty, \quad v = 0 \text{ at the supports} \right\}$$

and the middle condition — finite strain energy, since $v' = \varepsilon$ — is the one that does all the work. So the whole question reduces to:

for each function, is $\int (v')^2 \, \mathrm{d}x$ *finite?*

2 The two functions and their derivatives, side by side

The hat. Differentiate each branch of (1):

$$v'(x) = \begin{cases} +1, & x < 1, \\ \text{undefined}, & x = 1 \quad (\text{left says } +1, \text{ right says } -1), \\ -1, & x > 1. \end{cases} \tag{2}$$

It genuinely fails to exist — at exactly one point. But

$$\int_0^2 (v')^2 \, \mathrm{d}x = \int_0^1 1 \, \mathrm{d}x + \int_1^2 1 \, \mathrm{d}x = 2, \tag{3}$$

because a single point has zero length and contributes nothing to an integral. The value $v'(1)$ is never needed. Note also that v' stays **bounded**: it only ever takes the values ± 1.

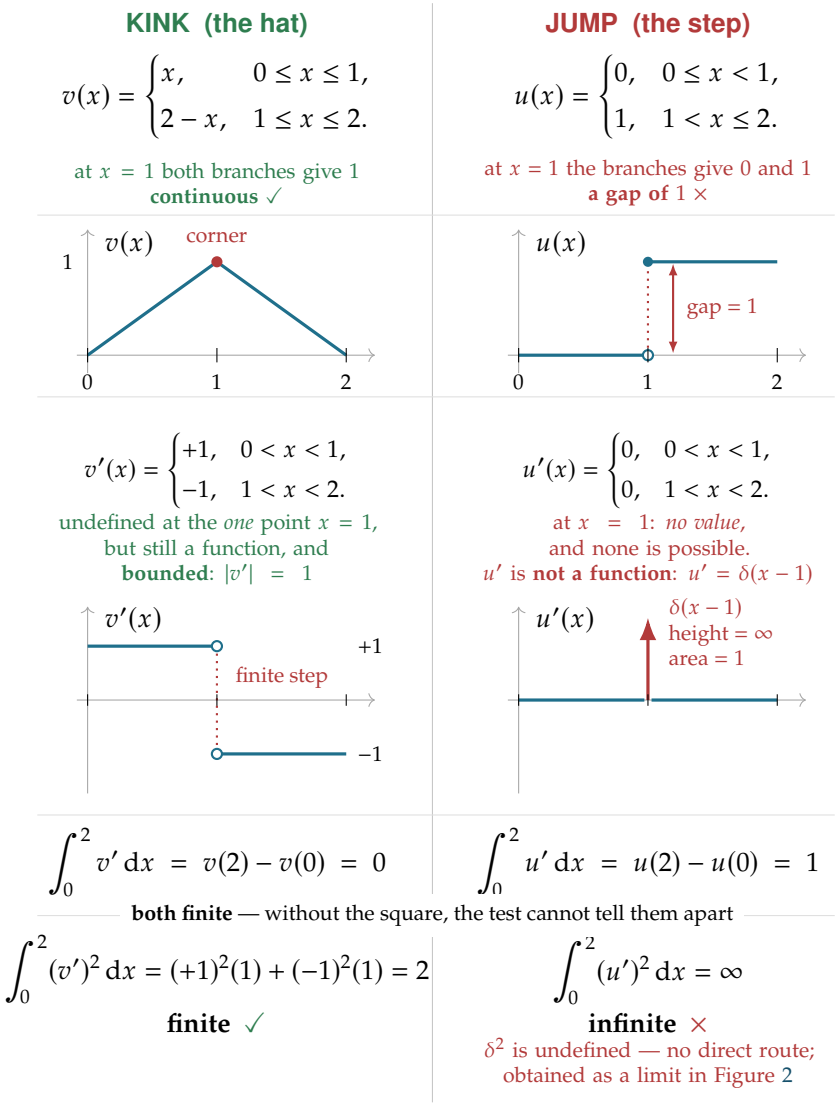


Figure 1: Rows: definition, shape, v' as a formula, v' plotted, $\int v'$, $\int (v')^2$. Only the last row separates the two cases.

The step. Away from $x = 1$ the function is flat, so $u' = 0$. At $x = 1$ it changes by 1 over a length of 0:

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{1}{0} = \infty.$$

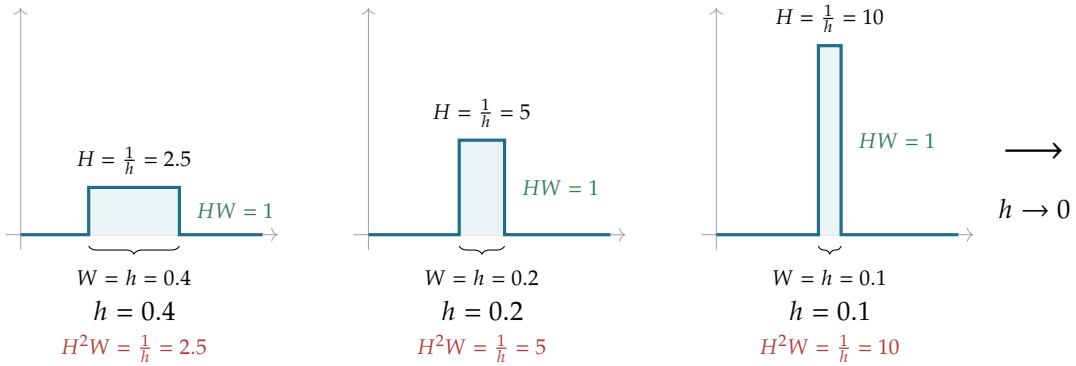
That is not a number, so guessing is no good. Instead, approach it.

3 Where the delta comes from: smooth the jump, then shrink

Replace the instantaneous jump by a steep ramp of width h centred at $x = 1$:

$$u_h(x) = \begin{cases} 0, & x \le 1 - \frac{h}{2}, \\ \dfrac{x - (1 - h/2)}{h}, & 1 - \frac{h}{2} \le x \le 1 + \frac{h}{2}, \\ 1, & x \ge 1 + \frac{h}{2}. \end{cases} \qquad \Longrightarrow \qquad u_h'(x) = \begin{cases} \dfrac{1}{h}, & \text{on the ramp,} \\ 0, & \text{elsewhere.} \end{cases} \tag{4}$$

So u_h' is a rectangle of height $1/h$ and width h. Now shrink h.



Where $1/h$ comes from. For a rectangle, squaring changes the *height* only — the width comes along untouched:

$$\int u_h' \, dx = H \cdot W = \tfrac{1}{h} \cdot h = 1 \qquad \text{a tie: the } h\text{'s cancel, so this is 1 for every } h$$

$$\int (u_h')^2 \, dx = H^2 \cdot W = \tfrac{1}{h^2} \cdot h = \tfrac{1}{h} \qquad \text{squaring beats the width: one } h \text{ survives below}$$

The height grows like $1/h$ and its square like $1/h^2$, but the width only shrinks like h. Squaring doubles the growth and leaves the shrinkage alone, so $H^2W = 1/h \to \infty$.

Figure 2: Shrinking h. Height $H = 1/h$, width $W = h$, so $HW = 1$ always — but $H^2W = 1/h \to \infty$.

h	height $H = \frac{1}{h}$	width $W = h$	$\int u_h' \, dx = HW$	$\int (u_h')^2 \, dx = H^2W$
0.1	10	0.1	1	10
0.01	100	0.01	1	100
0.001	1000	0.001	1	1000
$\to 0$	$\to \infty$	$\to 0$	**1**	$\to \infty$

Three things happen at once:

(1) the height $\to \infty$;

(2) the width $\to 0$;

(3) the **area stays exactly** 1 — always, since $(1/h) \cdot h = 1$.

Intuition: what a "delta spike" actually is

The limit is the **Dirac delta** $\delta(x-1)$: infinite height, zero width, area 1.
It is *not a function* — ordinary integration cannot see a single point, so no function can be infinite there and still enclose area 1. It is defined only by what it does inside an integral:

$$\int \delta(x-x_0)\,dx = 1 \qquad \text{(unit area)}, \qquad \int f(x)\,\delta(x-x_0)\,dx = f(x_0) \qquad \text{(samples } f \text{ at } x_0\text{)}.$$

Hence it is drawn as an **arrow**: the arrow's height on the page is its *area*, not its height. Height 3 means 3δ.
The derivative of a step is a delta.

Where this trips people up: never write "$u'(1) = \infty$" — it gives the wrong integral

It is tempting to record the spike as a pointwise value:

$$u'(x) = \begin{cases} 0, & x \neq 1, \\ \infty, & x = 1. \end{cases} \qquad \times \textbf{ This is not merely sloppy — it is false.}$$

Decisive test — FTC pins the answer at $\int_0^2 u'\,dx = u(2) - u(0) = 1$. But that formula is a genuine function, zero except at one point, and a single point has zero width, so it contributes nothing *whatever* value sits there (∞ included):

$$\int_0^2 u'\,dx = 0 \neq 1.$$

Writing ∞ destroys the one property that makes δ worth having.

Phrasing	Verdict	Why
$u'(1) = \infty$	× wrong	yields $\int u' = 0$ instead of 1
$u'(1)$ undefined, "1/0"	partly right	honest, and the "1" is really the area — but it is not a value, and it loses the *location*
$u'(1)$ has *no value*	✓ correct	accurate, though it sounds emptier than the truth

All three share the same false premise: u' is not a function, so it has no entries at points to fill in. The honest statement is two lines:

$$u'(x) = 0 \text{ for } x \neq 1, \qquad \text{and, as a distribution,} \qquad u' = \delta(x-1),$$

the second carrying what the first cannot — *where* (1) and *how much* (area 1). Keep "1/0" as scaffolding: a jump of size 3 gives 3/0 and yields $3\delta(x-1)$, so the numerator is the coefficient.

4 Why the kink stays tame: the same experiment on the hat

Do the identical smoothing to the hat's corner — let v_h' ramp linearly from +1 down to −1 across a window of width h.

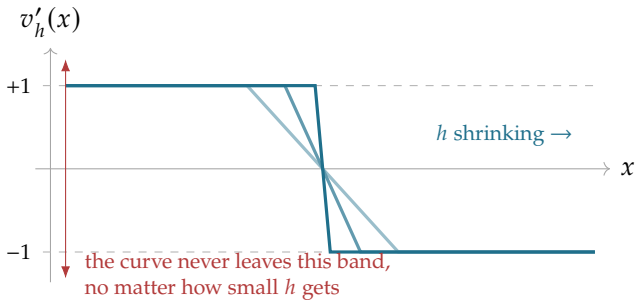


Figure 3: The same smoothing on the kink: v_h' never leaves $[-1, +1]$, so nothing can blow up.

h	$\max\lvert v_h'\rvert$	$\int_0^2 (v_h')^2\,\mathrm{d}x = 2 - \frac{2h}{3}$
0.1	1	1.933
0.01	1	1.993
0.001	1	1.9993
$\to 0$	**1**	$\to$ **2** (finite)

The height never grows. That single fact is the whole difference. Squaring a bounded function leaves a bounded function, and a bounded function on a bounded interval has a finite integral. Squaring a spike of height $1/h$ gives $1/h^2$, and multiplying by the width h still leaves $1/h \to \infty$.

5 Auditing the four integrals: when may you split at the break?

Every entry in the last two rows of Figure 1 was reached by the same instinct — *split the domain at $x = 1$, integrate each piece, add*:

$$\int_0^2 (\,\cdot\,)\,\mathrm{d}x \;\overset{?}{=}\; \int_0^1 (\,\cdot\,)\,\mathrm{d}x \;+\; \int_1^2 (\,\cdot\,)\,\mathrm{d}x \qquad \textit{the naive piecewise method.} \tag{5}$$

Run it on all four cells, checking each against something known independently. Two pass, two fail — along exactly the kink/jump line.

(1) $\int_0^2 v'\,\mathrm{d}x$ — PASS

On $(0, 1)$, v' is the constant $+1$ (antiderivative x); on $(1, 2)$ it is the constant -1 (antiderivative $-x$):

$$\int_0^1 (+1)\,\mathrm{d}x + \int_1^2 (-1)\,\mathrm{d}x = \big[x\big]_0^1 + \big[-x\big]_1^2 = (1-0) + \big((-2)-(-1)\big) = 1 - 1 = 0.$$

Independent check. The fundamental theorem gives $v(2) - v(0) = 0 - 0 = 0$. **Agrees.**

Why. The split discards $x = 1$; there v' is *bounded*, and zero width × bounded height contributes nothing.

(2) $\int_0^2 (v')^2\,\mathrm{d}x$ — PASS

Square first: $(+1)^2 = 1$ and $(-1)^2 = 1$, so *both* pieces have antiderivative x:

$$\int_0^1 (+1)^2\,\mathrm{d}x + \int_1^2 (-1)^2\,\mathrm{d}x = \big[x\big]_0^1 + \big[x\big]_1^2 = (1-0) + (2-1) = 1 + 1 = 2.$$

Independent check. $(v')^2 = 1$ at every point except $x = 1$, so the integrand is just the constant 1 and $\int_0^2 1\,dx = 2$. **Agrees.**

Why. Same reason — the square of a bounded function is still bounded.

Remark 5.1. Comparing (1) and (2): identical structure, and the only difference is the antiderivative of the second piece — $-x$ becomes $+x$, because $(-1)^2 = +1$. That single sign change turns cancellation ($1 - 1 = 0$) into accumulation ($1 + 1 = 2$).

(3) $\int_0^2 u'\,dx$ — FAIL

On both open pieces $u' = 0$:

$$\int_0^1 0\,dx + \int_1^2 0\,dx = 0 + 0 = 0.$$

Independent check. The fundamental theorem gives $u(2) - u(0) = 1 - 0 = 1$.

$$0 \neq 1 \qquad \times \textbf{ the method has lost the entire answer.}$$

Why. The split assumes nothing happens at $x = 1$ — true for v', false here. The pieces are empty; the whole answer sits at $x = 1$ as the unit area of $\delta(x - 1)$.

(4) $\int_0^2 (u')^2\,dx$ — FAIL

Squaring zero changes nothing, so the naive method again returns

$$\int_0^1 0\,dx + \int_1^2 0\,dx = 0.$$

Independent check. The smoothing of §3 gives $\int_0^2 (u_h')^2\,dx = 1/h \to \infty$.

$$0 \neq \infty \qquad \times$$

Why. Same illegal split; what sits at $x = 1$ is now unbounded, so discarding it turns ∞ into 0. And $(u')^2$ is not the derivative of anything, so there is no FTC fallback — smoothing is the *only* route here.

The scorecard

Integral	Naive gives	True value	Verdict	Checked against
$\int_0^2 v'\,dx$	0	0	pass ✓	FTC
$\int_0^2 (v')^2\,dx$	2	2	pass ✓	direct ($(v')^2 \equiv 1$)
$\int_0^2 u'\,dx$	0	1	fail ×	FTC
$\int_0^2 (u')^2\,dx$	0	∞	fail ×	the limit, Figure 2

Intuition: the pass/fail line *is* the kink/jump line

Splitting at $x = 1$ and integrating only the two open pieces silently discards the split point. So (5) is legitimate exactly when that point contributes nothing:

The hat (v' *bounded*)	**The step** (u' a *delta*)
zero width × height at most 1	the pieces are empty — *all*
= zero contribution	the content sits at $x = 1$
discarding it is harmless	discarding it discards everything
splitting is legal	**splitting is illegal**

The naive method passes on a kink and fails on a jump — for both integrals. Splitting at a break asserts that nothing lives there: true for a kink, false for a jump, and the method then returns a wrong answer silently.

6 The verdict

Translation: kink vs. jump, everything at once

	KINK (hat)	**JUMP** (step)
what is broken	the *slope* v'	the *shape* v itself
v is	continuous	discontinuous
v' near the break	*steps* $+1 \to -1$	*spikes* to $1/h$
height of v'	bounded (always 1)	unbounded ($\to \infty$)
v' is	a function	a delta — not a function
$\int v' \, dx$	0	1 (finite!)
$\int (v')^2 \, dx$	2 finite ✓	∞ ×
in V?	**yes**	**no**
mechanically	a **material interface** — steel glued to aluminium: displacement continuous, strain jumps because E jumps	a **broken bar** — a gap has opened; the structure has come apart

Where this trips people up: the jump is finite — the *energy* is what is infinite

Look at the two integral rows above. For the step,

$$\int u' \, dx = 1 \quad \text{(finite)}, \qquad \int (u')^2 \, dx = \infty.$$

The displacement jump itself is a perfectly ordinary number: the bar opened by 1. What diverges is the *square*. A delta has finite area but infinite squared area.
This is precisely why the admissibility test is written with a square. Had the condition been $\int v' \, dx < \infty$, the broken bar would have passed — and the theory would admit configurations in which the structure has come apart. The square is what makes V physical, because the square is the energy.

7 Climbing the ladder: v, v', v''

The rule "differentiating a step produces a delta" can be applied twice. Take the finite element hat — the same function, now extended by zero outside $[0, 2]$, as it actually appears in a mesh.

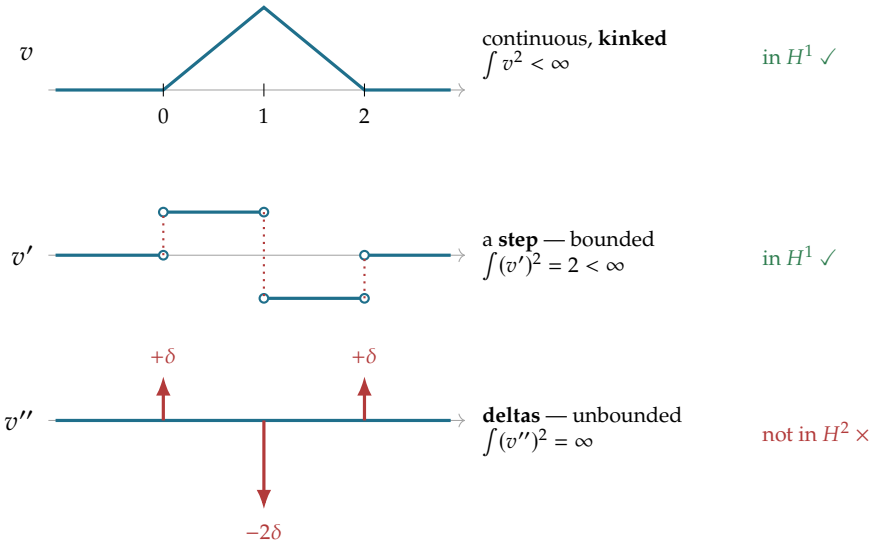


Figure 4: Differentiating the hat twice: kinked → bounded step → deltas. In H^1, not in H^2.

$$v'' = \delta(x) - 2\,\delta(x-1) + \delta(x-2). \tag{6}$$

The coefficients are just the jumps in v': it steps $0 \to +1$ at $x = 0$ (hence $+\delta$), then $+1 \to -1$ at $x = 1$ (a drop of 2, hence -2δ), then $-1 \to 0$ at $x = 2$ (hence $+\delta$).

Intuition: the hat is in H^1 but not in H^2

This settles the bar/beam question that otherwise looks like a separate rule.

- H^1 requires $\int (v')^2 < \infty$. The hat gives 2. **Pass** — so the hat is a legal *bar* displacement, and a legal FE shape function for a bar, a truss, a solid, or a heat conduction problem.
- H^2 requires $\int (v'')^2 < \infty$. The hat gives ∞. **Fail** — so the hat is *not* a legal *beam* deflection.

Physically: a hat-shaped beam has a corner, so its curvature v'' is a delta and $\frac{1}{2}\int EI(v'')^2 = \infty$. A hat-shaped beam is a **hinge**.

8 Reading the C^k labels off a polynomial

A single polynomial is C^∞. The classes appear only when you *glue* one to zero:

$$f(x) = \begin{cases} x^n, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad \text{on } [-1, 1]. \tag{7}$$

f $(x \geq 0)$	f	f'	f''	f'''	**class of** f
1	**JUMP**	—	—	—	C^{-1}
x	kink	**JUMP**	—	—	C^0
x^2	smooth	kink	**JUMP**	—	C^1
x^3	smooth	smooth	kink	**JUMP**	C^2
x^{k+1}	*the break marches one column right per power*				C^k

Rule. If $f^{(m)}$ is the first derivative to jump, then $f \in C^{m-1}$. So x^n glued to zero is exactly C^{n-1}: each extra power flattens the join by one more order.

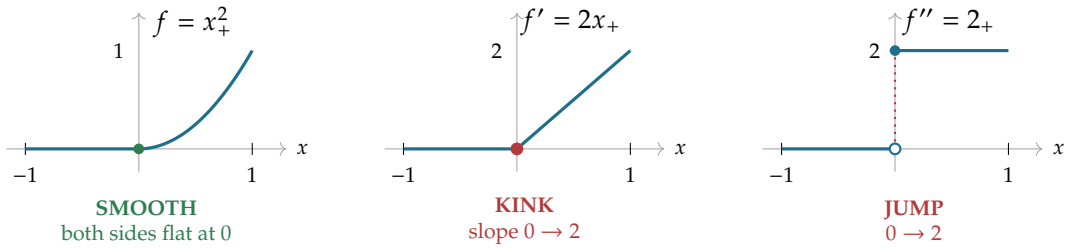


Figure 5: $f = x_+^2$ is exactly C^1: invisible in f, a kink in f', a jump only in f''.

Where this trips people up: "kink" and "jump" are relative — say *in what*

A kink in f *is* a jump in f' — one event, named one level apart.
Figure 5 is the case that catches people: $f = x_+^2$ looks perfectly smooth, because both sides are flat at 0. The defect cannot be seen in f at all; you must differentiate twice to reach it. Which is why beams are subtler than bars — a beam deflection can look smooth while its *curvature* w'' is the broken quantity.

9 What this decides in finite element analysis

Where each piece lands in practice.

9.1 Why the basis functions are hats in the first place

The chain runs backwards from what the method needs:

want	a sparse, banded stiffness matrix
⇒	basis functions with *small support* (each touches few neighbours)
⇒	the simplest such shape: the piecewise-linear **hat**
⇒	hats have *kinks*, so they are not C^1
⇒	the admissible space must **tolerate kinks**
⇒	use an *energy* (integral) test, not a *smoothness* (pointwise) test
⇒	$V = H_0^1$

H^1 was reverse-engineered to be exactly big enough to hold hats and small enough to exclude broken bars. Insist on classical differentiability and piecewise-linear FEM becomes illegal: you would need globally smooth basis functions, whose global support destroys sparsity.

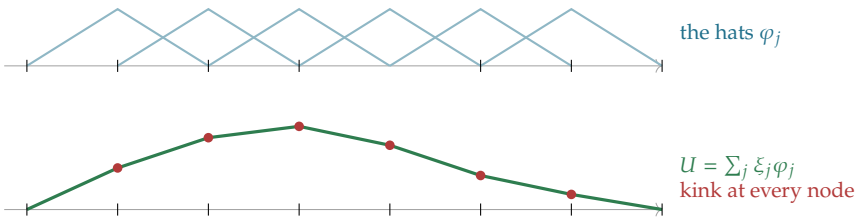


Figure 6: The FE solution is a sum of hats — a kink at every node, all legal. It must never *jump*.

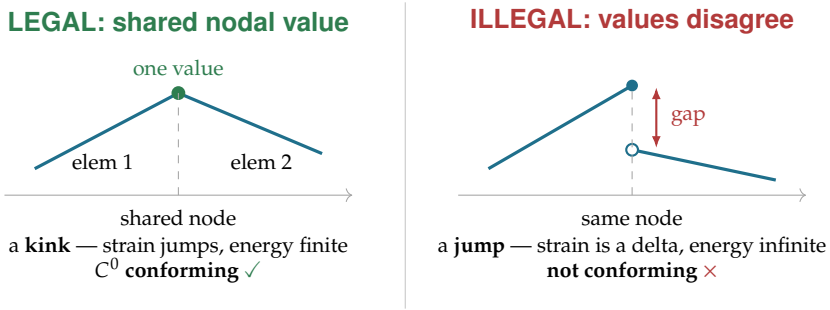


Figure 7: Sharing one nodal DOF is what forbids the right-hand gap, i.e. what enforces $V_h \subset V$.

9.2 The compatibility rule: what element assembly is really enforcing

So assembly is not bookkeeping: adding element contributions into shared rows and columns *is* the enforcement of $V_h \subset V$. A discrete space not inside V is *non-conforming* and needs its own convergence theory — the patch test polices exactly this.

9.3 Practical rule: put nodes where the kinks belong

The FE space can reproduce a kink **only at a node**. Inside an element, a linear shape function is perfectly straight and cannot bend. So if the true solution has a kink somewhere, the mesh must have a node there.

The most common case is a material interface.

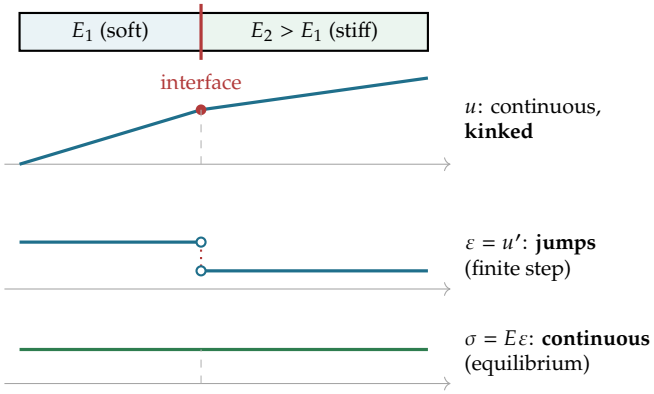


Figure 8: Bimaterial bar: σ continuous (equilibrium), so $\varepsilon = u'$ jumps and u kinks. **Put a node at the interface** — a linear element cannot bend inside itself.

Same for any abrupt change — a step in area, a point load, a change in distributed load, the edge of a contact patch. *Put a node there.*

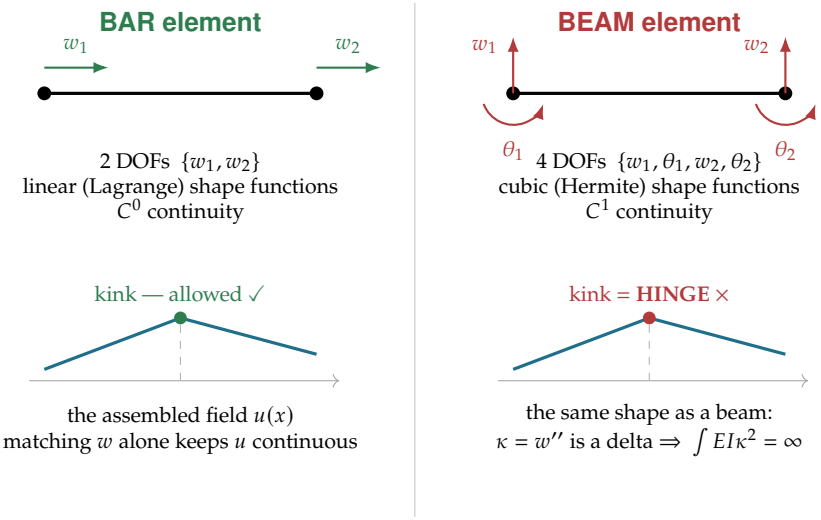


Figure 9: Bar: matching w keeps u continuous. Beam: the energy holds w'', so the *slope* must match too — else every node is a hinge.

9.4 Bars versus beams: the rule shifts up one rung

	Bar / truss / solid / heat	**Beam / plate / shell**
field	w = displacement	w = deflection
energy contains	w' (strain)	w'' (curvature)
PDE order	2 $\quad -(EA\,w')' = f$	4 $\quad (EI\,w'')'' = q$
space	H^1: $\int (w')^2 < \infty$	H^2: $\int (w'')^2 < \infty$
must stay continuous	w	w *and* w'
kink in w	allowed ✓	**forbidden** × (a hinge)
jump in w	forbidden ×	forbidden ×
element continuity	C^0	C^1
nodal DOFs	$\{w\}$	$\{w, \theta\}$
shape functions	linear hat (Lagrange)	cubic Hermite

The master rule. *The highest derivative appearing in the energy fixes the space, which fixes the continuity the elements must have, which fixes the degrees of freedom at each node.* Everything in the table follows from the left-hand column of its own row.

9.5 When you genuinely *want* a jump

Real structures do crack, delaminate, slip, and separate. Since a conforming C^0 space cannot represent a jump, you have to introduce one deliberately:

Situation	**How the jump is admitted**
crack along element boundaries	duplicate the nodes; the two faces get independent DOFs
crack cutting through elements	XFEM / GFEM: *enrich* the basis with a discontinuous function
progressive separation	cohesive-zone elements: a traction–separation law across the interface
contact / frictional slip	separate meshes plus constraints (multipliers, penalty)
shocks, advection dominance	discontinuous Galerkin: give up $V_h \subset V$ on purpose

None of these arise by accident: a conforming FE model *cannot* tear on its own. That guarantee is what makes ordinary structural analysis trustworthy, and the obstacle you must go around to model fracture.

Flip side: the FE solution is continuous with kinks at nodes, so its *stress is discontinuous across element boundaries*. The jumpy contour plot before smoothing is V_h being honest — and the size of those jumps measures discretisation error, the basis of the Zienkiewicz–Zhu estimator.

10 Summary

Intuition: the one-page version

A kink breaks the slope. A jump breaks the shape.

- Differentiating a kink gives a **step**: bounded, squares harmlessly, finite energy. *Allowed.*
- Differentiating a jump gives a **delta**: unbounded, finite area but infinite squared area, infinite energy. *Forbidden.*
- $V = H^1_0$ is exactly the set that admits the first and excludes the second — big enough for hat functions, small enough to keep the bar in one piece.
- Every FE displacement field is kinked at every node, by construction, and that is fine. What element assembly enforces is only that it never jumps.
- Differentiate once more and the hat's second derivative is a delta — so the hat is in H^1 but not H^2. That one fact separates C^0 bar elements from C^1 beam elements, and is why beam nodes carry rotations.
- Where the exact solution kinks (material interface, point load, section change), **put a node**.